



Permutation Riddle

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Difficulty Level : Medium

Permutation Riddle

Ramzy is tired of waiting for the student allowance and has started looking for a way to earn some money; luckily, he stumbled upon a strange man named "The Permutation Prince" who challenged him to a "Permutation Riddle".

"Find the number of permutations¹ of size n where neighbors differ by no more than d . Solve this count, and you shall claim a fortune equal to the lexicographically largest arrangement of six!"

Intrigued by the interesting challenge, and the large prize of 654,321 DA, Ramzy needs your help to solve this problem.

Task: Find the number of permutations of size n where each pair of adjacent elements has an absolute difference of at most d .

Input

The first and only line of input contains two integers n and d ($1 \leq n \leq 20, 1 \leq d \leq 5$).

Output

Print a single integer — the number of permutations of size n whose adjacent elements differ by at most d , modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 20$
- $1 \leq d \leq 5$
- Time limit: **2.0 seconds**
- Memory limit: **256 MB**

Sample Input 1

4 2

Sample Output 1

12

¹A permutation of size n is an arrangement of the integers from 1 to n where each number appears exactly once.

Explanation

There are 12 permutations of size 4 whose adjacent elements differ by no more than 2.

The permutation $[1, 2, 4, 3]$ is valid because:

- $|2 - 1| = 1 \leq 2$
- $|4 - 2| = 2 \leq 2$
- $|3 - 4| = 1 \leq 2$

However, the permutation $[1, 4, 2, 3]$ is **not** valid, since $|4 - 1| = 3$, which is greater than $d = 2$.

Sample Input 2

3 2

Sample Output 2

6